

Reaction and deactivation kinetics for Fischer–Tropsch synthesis on unpromoted and potassium-promoted iron catalysts

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Abstract

Rate data for Fischer–Tropsch synthesis on unsupported Fe and Fe/K catalysts were obtained at 10 atm total pressure, 200–240°C and H₂ and CO partial pressures of 0.9–1.4 and 0.4–0.9 atm, respectively. Based on a simple power-law expression, reaction orders with respect to H₂ and CO are 1.4 and –0.42 for the unpromoted Fe catalyst and 0.60 and –0.05 for the K-promoted Fe catalyst. A mechanistically-based rate model for FT synthesis is shown to be consistent with the kinetic data obtained for the Fe catalyst. Deactivation rate data were obtained during synthesis on the same catalysts at 10 atm total pressure, 250–280°C and H₂ and CO partial pressures of 0.6–1.4 and 0.3–0.9 atm, respectively, for periods of 24–65 h; from a fit of these data to a generalized power-law expression (GPLe) of the form $r_d = k_d'(a - a_\infty)^m$, values of the deactivation order m are found to be unity for both catalysts. Activation energies for deactivation are 108 and 58 kJ/mole for Fe and Fe/K catalysts, respectively. A deactivation model involving parallel paths is proposed, i.e. simultaneous conversion of (a) atomic to polymeric to graphitic carbon and (b) active carbon-rich carbides to inactive carbon-poor carbides. ©1999 Elsevier Science B.V. All rights reserved.

Keywords: CO hydrogenation; Catalyst deactivation; Fischer–Tropsch synthesis; Iron; Reaction kinetics; Deactivation kinetics

1. Introduction

Fischer–Tropsch synthesis (FTS) is an economically-promising, environmentally sound and technically-demonstrated technology for production of middle-distillate fuels, chemical feedstocks, and waxes from either coal or natural gas [1–4]. Considerable progress has been made in the past two decades in the development of more active, selective cobalt and iron catalysts and more effective reactor/process technologies [3,4]. Considerable research during this same period has focused on catalyst activity–structure relationships, reaction mechanisms and reaction ki-

netics [1–6] leading to a reasonably good understanding of these issues. At the same time, however, relatively little previous research has focused on the kinetics and mechanisms of catalyst deactivation. Yet, the most significant opportunities for improving the FTS process lie in the development of more stable, regenerable catalysts.

A principal cause of iron FTS catalysts deactivation is the transformation of active surface carbon species and/or active iron carbide phases to inactive carbon or carbide forms that foul or poison the surface [6–14]. However, previous work to characterize the carbonaceous species and mechanisms responsible for deactivation has been relatively limited [8–14]. While more than 10 different kinds of carbon-containing species have been observed to form during FTS on iron cat-

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alysts [8–14] i.e., atomic carbon, amorphous carbon, oxygenated carbon, polymeric condensed hydrocarbons, four different carbides, amorphous graphitic carbon, vermicular (filamentous) carbon, and crystalline graphitic carbon, only four of these, i.e., two less active graphites and two low-activity iron carbides are implicated in catalyst deactivation [8–14]. Other than our previous study of the kinetics of deactivation of iron catalysts by carbon at low reaction pressure (1 atm) [15] no studies of deactivation kinetics have been reported.

This work focused on the measurement of the kinetics of deactivation of Fe and Fe/K catalysts by carbon at high-pressure reaction conditions, i.e. 10 atm and 200–250°C. However, since the complete set of equations describing deactivation kinetics includes a rate equation for the main reaction, it was necessary to measure for these same Fe and Fe/K catalysts reaction kinetics for the main reaction under conditions for which the deactivation rate was negligible. While reaction kinetics have been reported for unpromoted and potassium-promoted Fe Fischer–Tropsch catalysts [15–21], the catalysts and/or conditions used in these previous studies were sufficiently different to rule-out combining the reported kinetic expressions with our deactivation rate data. Hence, it was necessary to obtain reaction parameters over a wide range of conditions on our catalysts to meet the needs of this study. These parameters were obtained under conditions of little or no deactivation as well as under conditions for which deactivation plays a major role so that effects due to reaction and deactivation could be separated.

2. Experimental

2.1. Catalyst preparation and characterization

The Fe catalyst was prepared by thermal decomposition of reagent grade $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (1 wt.% Al/Fe) in air at 120°C until dry followed by calcination at 200°C for 12 h. The Fe/K catalyst was prepared by adding KNO_3 dissolved in water to a reduced and passivated sample of the Fe catalyst to obtain a total molar K/Fe ratio of 0.013. Each catalyst was reduced at 400°C in flowing hydrogen before use.

Catalysts samples were prepared for activity/kinetic tests by wash-coating the previously reduced and passivated catalyst onto 6.25 cm long $\times$ 2.5 cm diameter ceramic monoliths (Corning Celcor) having a cell density of 300 square channels per square inch by the following procedure: (1) the catalyst was crushed to a fine powder with mortar and pestle and sieved through a 50-mesh screen to eliminate large particles that might significantly block the monolith channels; (2) a high-surface-area alumina binder (Conoco Dispal) was physically mixed with either the Fe or Fe/K catalyst in a 30/70 weight ratio; (3) the solid mixture was then slurried in water and wash-coated onto the surfaces of monolith channels by pouring the slurry through the channels. Monoliths were dried at 200°C for 1 h between each of the several coats and after completion of the wash-coating procedure.

Static hydrogen chemisorptions were performed using a standard volumetric glass adsorption apparatus by the following procedure: (1) catalysts reduced at 400°C were evacuated for 1 h at 400°C and cooled to 100°C in vacuum and (2) total H_2 uptake was determined at 100°C by measuring uptake as a function of pressure over the range of 100–400 Torr and extrapolating the straight-line portion of the isotherm to zero pressure. The H_2 uptake of iron catalysts at 100°C was shown in earlier studies to provide an accurate measure of the adsorbed monolayer corresponding to a H/Fe ratio of unity, while the extrapolation to zero pressure eliminates contributions to physical adsorption on the support [22].

2.2. Reactor apparatus and activity-testing procedure

Activity and kinetic tests were conducted on catalysts in the form of washed-coated monoliths to (1) minimize pressure drop in our recycle CSTR reactor and (2) minimize pore-diffusional resistance. Activity and selectivity measurements were performed in a Berty fixed-catalyst-bed, internal-gas-recycle reactor at 10 atm total pressure, 200–280°C (200–240°C for intrinsic kinetic measurements and 250–280°C for deactivation rate measurements), H_2/CO ratios of 1–3 with H_2 and CO partial pressures (in the feed) ranging from 1 to 1.5 and from 0.5 to 1.0 atm, respectively, (the remainder He diluent) and CO conversions of less than 10% for kinetic measurements and less than 40% dur-

ing deactivation experiments; space velocities (based on a monolith volume of 32 cm³) were in the range of 400–700 h^{−1}. Thus, partial pressures of H₂ and CO in the reactor were 0.9–1.4 and 0.4–0.9 for kinetic studies and 0.6–1.4 and 0.3–0.9 for deactivation studies.

Fe or Fe/K catalyst (1.5 g), washcoated on a 2.5 cm diameter × 6.25 cm long monolith (48 channels/cm²) was charged to the reactor. Reactant gases (H₂, CO and He as diluent) were purified separately before mixing by passing them through deoxo and zeolite guard beds. Reaction products were passed directly from the reactor through heated lines to a Hewlett Packard gas chromatograph (5890A) for analysis with thermal conductivity and flame-ionization detectors. Temperature programming (−50°C to 235°C) with a packed Chromosorb-102 column and a capillary column allowed identification of non-condensable gases, water and light hydrocarbons (C₁ to C₁₅). A microcomputer used to perform data acquisition and process control also controlled automatic chromatograph sampling.

For reaction kinetic studies, each catalyst was initially operated under synthesis conditions for a period of 20–24 h before collection of rate data to ensure that ‘steady-state’ conditions were achieved with respect to the C₁ to C₁₂ hydrocarbons. Subsequent changes in temperature or partial pressure were followed by reaction for 6–12 h before collection of rate data. Multiple samples were analyzed at each steady-state condition. Sampling for the deactivation studies began approximately 20 min after the introduction of reactants into the catalytic chamber. The sampling frequency was decreased from once each hour during the initial 12 h, involving relatively rapid activity loss, to once each 3 h beyond 12 h of reaction time.

3. Results

3.1. Chemisorption measurements

Hydrogen chemisorption uptakes of 70 and 50 μmoles H₂/g were obtained for the fresh Fe and Fe/K catalysts, respectively.

3.2. Kinetic and selectivity data in the absence of deactivation

Rate data obtained for the Fe and Fe/K catalysts over a range of temperatures (200–240°C) and H₂/CO

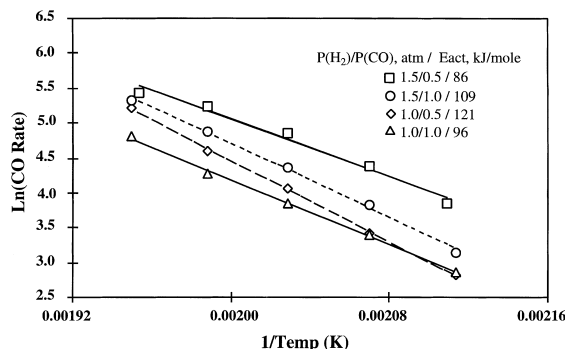


Fig. 1. Apparent activation energies for monolith-supported, unpromoted iron (Fe) at 10 atm and different feed gas concentrations.

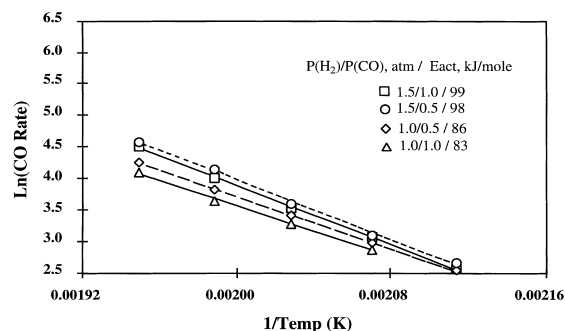


Fig. 2. Apparent activation energies for monolith-supported, K-promoted iron (Fe/K) at 10 atm and different feed gas concentrations.

ratios (1–3) are listed in Table 1 and plotted in Arrhenius form in Figs. 1 and 2. Standard deviations from the mean for these data obtained from two different gas samples for each condition, range from 1% to 10%. Activation energies based on a regression of all the data in Figs. 1 and 2 are 101 ± 4.4 and 92 ± 3.1 kJ/mol for Fe and Fe/K catalysts, respectively (Table 2). Rate data from reaction at different H₂/CO ratios and temperatures were regressed using the power-law expression in Eq. (1) to give the exponents listed in Table 2. The values of x and y (orders for H₂ and CO) for the Fe catalyst are significantly higher and lower, respectively, relative to those for Fe/K.

$$r = k (P_{\text{H}_2})^x (P_{\text{CO}})^y \quad (1)$$

Representative reactant and product partial pressures and product selectivities obtained during kinetic runs at 10 atm and after 9 h of reaction, a time adequate for reaching a near steady-state, are listed in Table 3. Under these low conversion conditions, partial

Table 1

CO conversion rates for consecutive runs (same sample) of Fe and Fe/K catalysts at 200–240°C and H₂/CO = 1–1.5 after 6 h of reaction at each given condition

Catalyst	Temperature (°C)	H ₂ /CO ^a ratio	CO conversion ^b %	CO rate ^c (μmole/g-min)	Standard deviation ^d (%)
Fe catalyst ^e (unpromoted, monolith-supported)	200	1.5	0.47; 0.44	18.2; 17.2	4.0
	210	1.5	1.2; 1.2	46.1; 45.4	1.1
	220	1.5	2.3; 2.0	90.0; 78.3	9.8
	230	1.5	3.5; 3.4	136; 132	2.1
	240	1.5	5.4; 5.3	208; 204	2.6
Fe/K catalyst ^f (K-promoted, monolith-supported)	210	1.0	1.4; 1.4	60; 57	2.9
	220	1.0	2.2; 2.1	91.7; 89.7	1.6
	230	1.0	3.3; 2.9	138; 121	9.3
	240	1.0	4.6; 5.0	192; 211	6.7

^a Molar ratio of feed; CO flow of 150 sscm; $P_{CO}^0 = 1$ atm.

^b CO conversion (X_{CO}) = $(v^0 V_{CO}^0 - v^{exit} V_{CO}^{exit}) / v^0 V_{CO}^0$ where V_{CO} is the molar volume (μmol/cc) and v^0 is the volumetric flow rate in cc/min.

^c CO rate (μmol/g-min) = (CO conversion) × [CO Flow (cc/min)] × [molar vol. @ 25°C (μmol/cc)] / catalyst charge (g) or CO rate = $X_{CO} \times V_{CO} \times 40.89 / W_{cat}$.

^d Standard deviation divided by the mean times 100.

^e 99% Fe/1% Al₂O₃.

^f 97.7% Fe/1% Al₂O₃/1.3% K.

Table 2

Activation energies and reaction orders for H₂ and CO for CO conversion during CO hydrogenation on Fe and Fe/K catalysts at $P_{CO}^0 = 0.5$ –1.0 and $P_{H_2}^0 = 1.0$ –1.5 atm and $X_{CO} < 0.10$

Catalyst and conditions	H ₂ order (X)	CO order (Y)	Activation energy (kJ/mole)	
			Regression	Range
Monolith-supported, unpromoted Fe	1.4 (±0.15)	−0.42 (±0.09)	101 (±4.4)	86–121
Monolith-supported, K-promoted Fe	0.60 (±0.10)	−0.05 (±0.06)	92 (±3.1)	83–99

Table 3

Representative reactant and product partial pressures and product selectivities during kinetic runs at 10 atm after 9 h reaction

Catalyst	T (°C)	H ₂ /CO	CO conversion (%)	H ₂ conversion (%)	P_{CO}^0 (atm)	$P_{H_2}^0$ (atm)	$P_{CO_2}^a$ (atm)	$P_{H_2}^b$ (atm)	$P_{H_2O}^c$ (atm)	$P_{H_2O}^d$ (atm)	$S_{CO_2}^e$	$S_{H_2O}^f$	S_{HC}^g
Fe	220	2	2.98	3.08	0.50	1.00	0.485	0.97	0.001	0.012	0.08	0.40	0.91
	240	2	9.57	9.34	0.50	1.00	0.452	0.90	0.005	0.039	0.09	0.41	0.90
Fe/K	220	2	3.97	4.83	0.50	1.00	0.480	0.95	0.002	0.026	0.12	0.53	0.87
	240	2	9.67	8.36	0.50	1.00	0.452	0.91	0.007	0.030	0.13	0.35	0.86
Fe	220	1	1.20	1.94	1.00	1.00	0.988	0.98	0.003	0.006	0.25	0.30	0.74
	240	1	3.16	5.96	1.00	1.00	0.968	0.94	0.005	0.022	0.16	0.36	0.84
Fe/K	220	1	2.14	4.39	1.00	1.00	0.979	0.95	0.003	0.021	0.12	0.48	0.87
	240	1	5.04	10.28	1.00	1.00	0.950	0.89	0.005	0.048	0.09	0.46	0.90

^a $P_{CO} = P_{CO}^0 (1 - X_{CO})$ (atm); contraction is assumed negligible (Typical value of −0.005).

^b $P_{H_2} = P_{H_2}^0 (1 - X_{H_2})$.

^c $P_{CO_2} = S_{CO_2} \times X_{CO} \times P_{CO}^0$.

^d $P_{H_2O} = S_{H_2O} \times X_{H_2} \times P_{H_2}^0$.

^e $S_{CO_2} = P_{CO_2} / (P_{CO}^0 \times X_{CO})$, at.% C.

^f $S_{H_2O} = P_{H_2O} / (P_{H_2}^0 \times X_{H_2})$ (mole fraction).

^g $S_{HC} = 1 - S_{CO_2}$ at.% C.

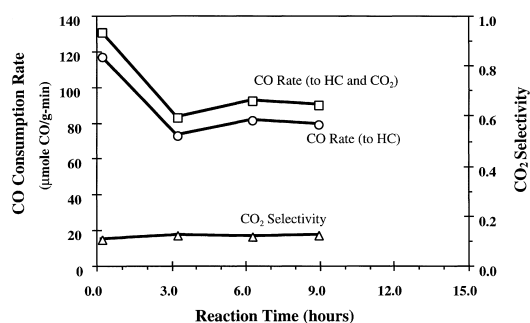


Fig. 3. Rates of CO conversion to all products or to hydrocarbons only and CO₂ selectivity for monolith-supported Fe/K catalyst at 220°C ($P_{\text{tot}} = 10$ atm, $P_{\text{H}_2}^0 = 1.0$ atm, $P_{\text{CO}}^0 = 1.0$ atm).

pressures of H₂ and CO are near (within 5–10% of) the values of feed, while partial pressures of CO₂ and H₂O are small, i.e., less than 1% and 10% of reactant partial pressures, respectively. Selectivities for CO₂ and H₂O based on CO and H₂ conversions, respectively, are about 10–14 and 40–54 mol%, while selectivities of CO for hydrocarbons are 86–90% (at.% C). Under these low-temperature, low-conversion conditions, the water–gas-shift reaction is apparently far from equilibrium.

Rates of CO disappearance at 220°C and (H_2/CO)_{feed} = 1 to either CO₂ and hydrocarbons or to hydrocarbons only, along with CO₂ selectivities are shown as a function of reaction time for Fe/K in Fig. 3. At these low-temperature reaction conditions near steady-state rates are observed after about 6 h. It is evident that CO₂ selectivity reaches steady state very quickly and remains at about 12% (at.% C from CO); thus, selectivity of CO to hydrocarbons is about 88% (at.% C). Very similar rate and selectivity versus time behavior was observed for Fe.

Typical hydrocarbon product distributions in mole% (and wt.%) of the hydrocarbon product and values of α (propagation probability) are listed in Table 4 at 240°C, 10 atm, and a H₂/CO ratio of 1. Under these conditions, mole fractions of methane and light hydrocarbons are high for both catalysts; mole fractions of C₅–C₁₁ liquids are significant. However, the methane make of the Fe/K catalyst is substantially lower (39 relative to 61 mol% or 11 versus 30 wt.%), its selectivity for C₅–C₁₁ liquids substantially higher (22 relative to 8 mol% or 51 versus 31 wt.%), its selectivity for C₁₂₊ liquids and waxes much higher (12 versus 3 wt.%), and its α value significantly higher (0.79 rel-

ative to 0.69) than for the potassium-free Fe catalyst; these differences are consistent with previously reported effects of potassium promotion.

In a series of tests on both Fe and Fe/K catalysts at temperatures in the range of 250–280°C and H₂/CO ratios of 1–3, α values ranged from 0.63 to 0.76 for Fe and 0.72 to 0.89 for Fe/K; α was found to decrease with increasing temperature and H₂/CO ratio for both catalysts. C₃–C₇ olefin selectivities were also measured over the same range of conditions; selectivities (mol% of C₃–C₇ products) were 29–84% for Fe and 27–81% for Fe/K; olefin selectivity was observed to decrease with increasing temperature and increasing H₂/CO ratio and was higher for Fe/K relative to Fe. In the same series of tests, selectivities of H₂ to H₂O for Fe and Fe/K catalysts were found to be 9–44 and 6–60 mol%, respectively; selectivities for CO to CO₂ for Fe and Fe/K catalysts were 17–30 and 9–26%, respectively. Values of the equilibrium constant for the water–gas-shift (WGS) reaction, $K = P_{\text{CO}_2} \times P_{\text{H}_2\text{O}} / (P_{\text{CO}} \times P_{\text{H}_2})$ were calculated from the measured partial pressure data; maximum values at any given temperature and H₂/CO ratio are listed in Table 5 and compared with theoretical values of K for each temperature. It is evident that the experimental partial pressure ratio is 5–10 times smaller than the predicted equilibrium constant at any given temperature and H₂/CO ratio, indicating that the WGS reaction operates far from equilibrium even at these relatively high reaction temperatures, as a consequence of the relatively low conversions and relatively low catalytic activity for the WGS reaction.

3.3. Deactivation rate data

Reaction rate versus time data are shown for CO hydrogenation on Fe and Fe/K catalysts at relatively high reaction temperatures in Figs. 4–7. Fig. 4 shows both CO and H₂ conversion rates for unpromoted Fe. Rates are expressed as the number of μmoles of CO (or H₂) reacted to produce CO₂, H₂O, and hydrocarbons per gram of catalyst per minute. Decreases in rate with time are similar for CO and H₂, namely an exponential decrease at early reaction times and an approach to an asymptotic value at long reaction times, although the rate of hydrogen consumption is consistently higher than CO consumption. Since, in principle, both CO

Table 4

Product distributions for Fe and Fe/K catalysts at 240°C, 10 atm and H₂/CO = 1

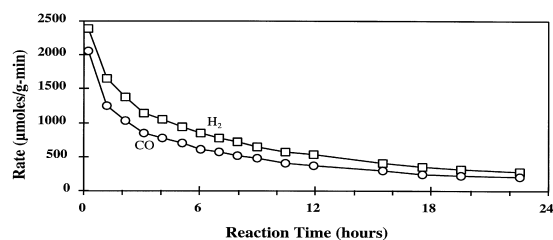
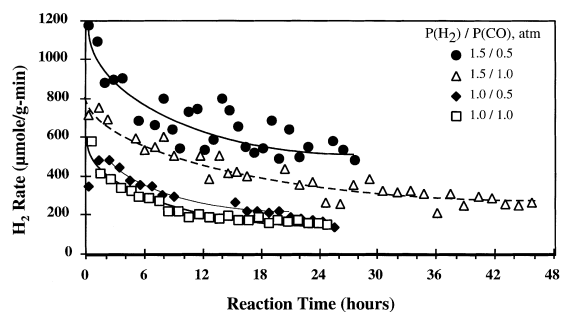
Catalyst	CO conversion (%)	Mole% (Wt.%) of hydrocarbon product										α^a
		C ₁	C ₂	C ₂ =	C ₃	C ₃ =	C ₄	C ₄ =	C _{5–11}	C ₁₂₊		
Monolith-supported Fe (10 atm, 240°C, H ₂ /CO = 1)	3.2	61 (30)	13 (12)	2 (2)	5 (7)	5 (6)	2 (4)	3 (5)	8 (31)	0.5 (3)	0.69	
Monolith-supported, K-promoted Fe (10 atm, 240°C, H ₂ /CO = 1)	4.6	39 (11)	13 (7)	3 (2)	5 (4)	7 (5)	2 (2)	5 (5)	22 (51)	3 (12)	0.79	

^a Propagation probability obtained by plotting log of mole fraction versus carbon number.

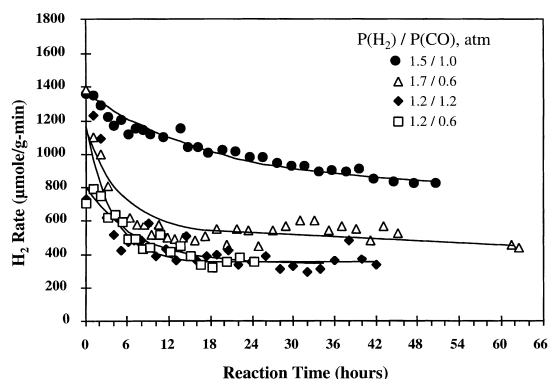
Table 5

Maximum values of the water–gas shift equilibrium constant calculated for Fe and Fe/K catalysts at various temperatures from measured partial pressure data (for which $P_{\text{H}_2}^0 = 1.2$ atm and $P_{\text{CO}}^0 = 0.8$ atm) and at different H₂/CO ratios (at $T = 260^\circ\text{C}$)^a

Catalyst	Temperature ($P_{\text{H}_2}^0 = 1.2$ atm; $P_{\text{CO}}^0 = 0.8$ atm)				H ₂ /CO ratio ($T = 260^\circ\text{C}$)			
	250°C	260°C	270°C	280°C	3	2	1.5	1
Unpromoted	2.8	1.4	3.1	1.8	10	0.96	1.9	–
K-promoted	0.70	0.42	0.74	1.7	3.4	2.6	1.1	0.52
Equilibrium	67	58	50	43	58	58	58	58

^a These data are compared with predicted values of the equilibrium constant at equilibrium.Fig. 4. Comparison of reaction rates of hydrogen and carbon monoxide on unpromoted Fe (280°C, $P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm).Fig. 5. Deactivation rates on unpromoted Fe at 260°C, 10 atm total pressure and different H₂ and CO partial pressures of the feed.

and H₂ rates follow the same trend, either could be used to assess deactivation kinetics. We have chosen to use the reaction rates based on H₂ disappearance

Fig. 6. Deactivation rates on Fe/K at 260°C, 10 atm total pressure and different H₂ and CO partial pressures of the feed.

for this purpose, because they could be fitted to deactivation functions with a greater statistical confidence; thus, rates are presented henceforth in this paper in terms of hydrogen consumption, except where the CO rate provides additional insight or is used simply for comparison purposes.

3.3.1. Variation in deactivation rate with reactant concentrations at constant temperature

Data from deactivation experiments on unpromoted (potassium-free) catalysts at a constant temperature of 260°C are shown in Fig. 5. All four rate versus time

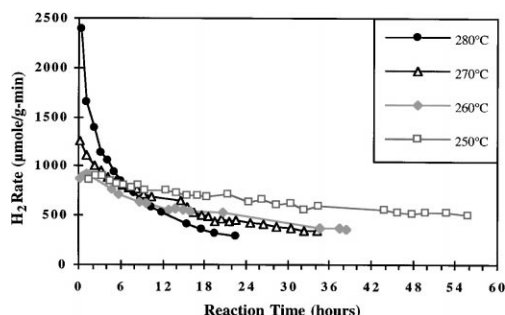


Fig. 7. Deactivation rates on unpromoted Fe at $P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm, 10 atm total pressure and reaction temperatures of 250–280°C.

curves have a very similar shape and show qualitatively the same trend as in Fig. 4. Thus, it appears initial rate depends strongly on reactant concentrations, while decreases in rate with time are fairly concentration independent. The highest rate is observed with the combination of the highest hydrogen partial pressure (1.5 atm) and the lowest carbon monoxide partial pressure (0.5 atm). Increasing the CO partial pressure from 0.5 to 1.0 atm decreases the rate of H_2 consumption by about a third. However, if hydrogen pressure is reduced to 1.0 atm, the same change in CO partial pressure effects only a small change in activity. Comparing runs in which the feed CO partial pressure is constant at 0.5 atm and H_2 feed partial pressures are 1.5 or 1.0 atm (i.e., the filled symbols), rates at the higher hydrogen pressure are nearly three times those at the lower pressure. With the CO pressure constant at 1.0 atm (open symbols), a decrease in the H_2 partial pressure from 1.5 to 1.0 atm also leads to a 50% decrease in H_2 consumption. The trends observed here are consistent with the trends from the kinetic data (i.e., obtained in the absence of deactivation): an increase in H_2 pressure leads to increased rates while an increase in CO pressure leads to decreased rates.

Observed rates of H_2 consumption over Fe/K at 260°C and different reactant partial pressures are shown in Fig. 6. However, even more than with the unpromoted Fe catalyst, the effects of H_2 and CO partial pressures are complex, and there are no consistent trends for initial rate with changes in initial reactant corrections. That the initial activities of the samples corresponding to the first two data sets are nearly twice those of the other samples is unexpected, given the CO and H_2 dependencies discussed earlier.

These differences may be due in part to differences in preparation of the wash-coated monolith used for each condition.

3.3.2. Effect of temperature on deactivation rate (reactant concentrations constant)

H_2 consumption rates of Fe and Fe/K catalysts were measured at temperatures ranging from 250°C to 280°C at constant H_2 and CO partial pressures in the feed of 1.2 and 0.8 atm, respectively. As seen in Fig. 7, initial H_2 conversion rates for unpromoted Fe are higher at the higher temperatures, i.e., increasing from 850 $\mu\text{mol/g-min}$ at 260°C to 2400 $\mu\text{mol/g-min}$ at 280°C. However, after reaction for 24 h, rates are lower at the higher temperatures because of more rapid deactivation, i.e. rates decrease in the order 250°C, 260°C, 270°C, 280°C. On the K-promoted catalyst, the highest initial rate was also observed at 280°C, while rates after 24 h were found to decrease in the order 250°C > 260°C > 280°C > 270°C (data are not shown in this case).

Effects of temperature on activity loss can be quantitatively compared when rates are plotted as normalized activity, a (i.e., rate at time t divided by initial rate, for a given temperature). Normalized activity-time plots are provided in Figs. 8 and 9 for unpromoted and potassium-promoted catalysts. It is evident from these plots that the extent of deactivation increases as the temperature increases. For example, at 250°C, the activity of the unpromoted Fe catalyst decreases only

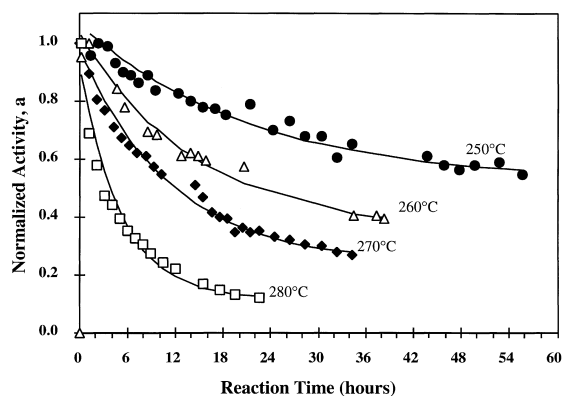


Fig. 8. Normalized activity of unpromoted Fe at $P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm, 10 atm total pressure and reaction temperatures of 250–280°C.

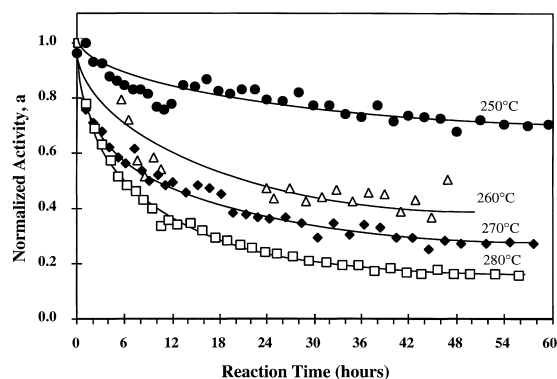


Fig. 9. Normalized activity of Fe/K at $P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm, 10 atm total pressure and reaction temperatures of 250–280°C.

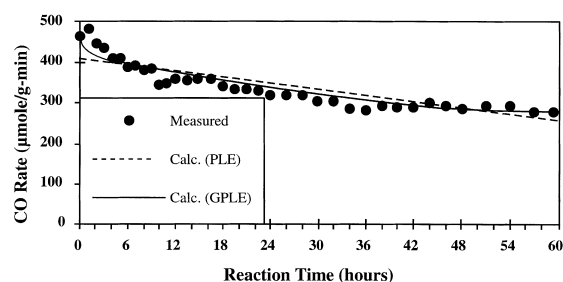


Fig. 10. Comparison of calculated CO conversion rates based on PLE and GPLE with measured rates for K-promoted iron (250°C, $P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm, $P_{\text{TOT}} = 10$ atm).

30% in 24 h, while it decreases 90% at 280°C over the same time period.

A generalized power-law expression (GPLE) is the appropriate rate form for deactivation phenomena in which normalized activity approaches an asymptotic value at long times [23],

$$r_d = k_d (P_{\text{H}_2})^x (P_{\text{CO}})^y (a - a_\infty)^m \quad (2)$$

where k_d refers to the deactivation rate constant, a is the activity, and a_∞ is the activity at infinite reaction time. Since the dependence of deactivation rate on H_2 and CO partial pressures was observed to be relatively secondary compared to the dependencies on activity and temperature, the data were fitted to a simplified, concentration-independent GPLE:

$$r_d = k'_d (a - a_\infty)^m \quad (3)$$

A deactivation order m of unity was found to fit the data well for both catalysts. Fig. 10 shows that this

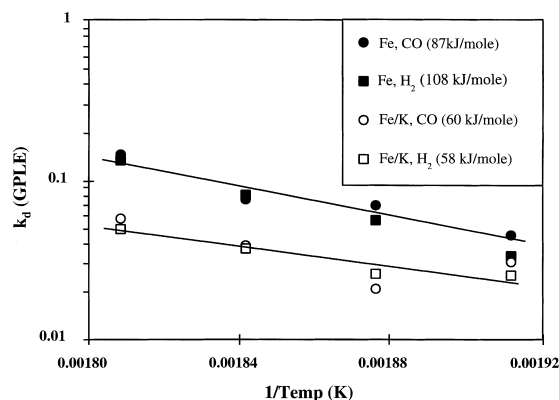


Fig. 11. Arrhenius plot of deactivation rate constants (k_d values) for unpromoted and potassium-promoted catalysts using the GPLe.

Table 6

Activation energies (kJ/mole) of deactivation for the unpromoted and K-promoted catalysts based on conversion rates of both CO and H_2 ($\text{H}_2/\text{CO} = 1.5$) based on the generalized power law

Catalyst	Activation energies of deactivation (kJ/mole)	
	CO	H_2
Unpromoted Fe	87	108
Fe/K	60	58

GPLe fits the data much better than a simple power law expression (PLE) in which the a_∞ term is omitted.

Values of the deactivation rate constant k_d are plotted in Arrhenius form in Fig. 11 from which activation energies were determined (Table 6). Values of 108 and 58 kJ/mole are obtained for Fe and Fe/K, respectively, based on H_2 conversion data. It is evident that deactivation rates (Fig. 11) and activation energies (Table 6) for deactivation are significantly larger for Fe relative to the Fe/K catalyst. However, deactivation rates and activation energies for deactivation are very significant for both catalysts. For example, 50% loss of activity is predicted at 260°C and a total pressure of 10 atm ($P_{\text{H}_2}^0 + P_{\text{CO}}^0 = 2.2$ atm, $\text{H}_2/\text{CO} = 1.5$) for Fe and Fe/K catalysts at 24 and 60 h of reaction, respectively. At temperatures ranging from 260°C to 280°C ($P_{\text{H}_2}^0 = 1.2$ atm, $P_{\text{CO}}^0 = 0.8$ atm) times for reaching 50% of initial activity are 6–24 h for the unpromoted iron and 19–60 h for the K-promoted iron (see Table 7). The relatively good linear correlation of the data in Fig. 11 further supports the use of simplified GPLe (Eq. (3)).

Table 7
Time (hours) to reach 50% of initial activity^{a,b}

T (°C)	Based on H ₂ <i>k_d</i> values ^b		Based on CO <i>k_d</i> values ^b	
	Unpromoted Fe	Fe/K	Unpromoted Fe	Fe/K
250	84	c	c	c
260	24	60	20	33
270	12	30	11	22

^a Data are based on the expression $a = (a_0 - a_\infty) e^{-k_d t} + a_\infty$ solved for t where a_0 is the initial activity (defined as 1.0) and a is the activity at time t ($a = a_0/2$ for this data set) and using regressed values of k_d at each temperature.

^b k_d values used are based on data at $P_{H_2}^0 = 1.2$ and $P_{CO}^0 = 0.8$ atm.

^c Since $a_{inf} > 0.5$, activity does theoretically reach half the initial value.

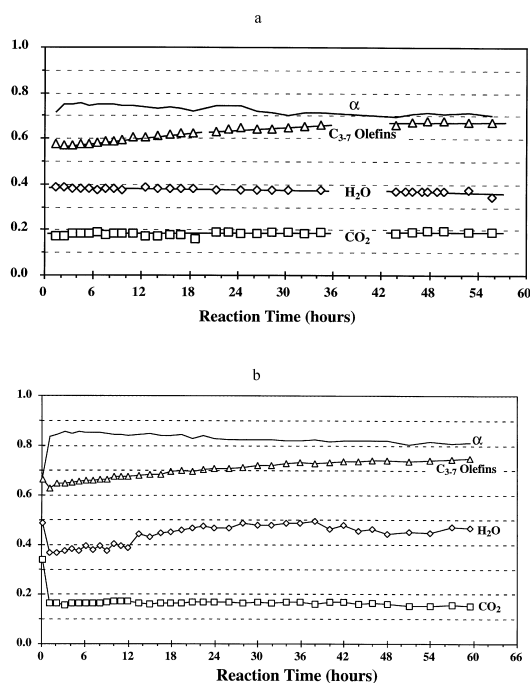


Fig. 12. Selectivity parameters for FT synthesis at 250°C, $P_{H_2}^0 = 1.2$ and $P_{CO}^0 = 0.8$ atm on (a) Fe and (b) Fe/K catalysts.

3.3.3. Effects of deactivation on product selectivity

CO₂, H₂O, and C_{3–7} olefin selectivities and values of the propagation probability α measured at 250°C and H₂/CO = 1.5 are plotted as a function of time for Fe and Fe/K catalysts in Fig. 12. All these selectivity parameters reach steady-state values in less than 2 h under these conditions. Over the subsequent 50–55 h, CO₂ selectivity is constant and α steadily decreases about 5–6% for both catalysts, while C_{3–7} olefin se-

lectivity increases 25% and 15% for Fe and Fe/K catalysts, respectively; H₂O selectivity is constant for Fe but increases about 20% for Fe/K.

4. Discussion

4.1. Reaction kinetics

4.1.1. Comparison of rate parameters from this and previous studies

Kinetics of CO hydrogenation on unpromoted and K-promoted Fe have been studied over a range of reaction conditions [15–21]. Reaction rates have generally been modeled using a simplified power-law expression (SPLE) of the form

$$-r_{CO} = k (P_{H_2})^X (P_{CO})^Y \quad (4)$$

Table 8 lists calculated X and Y parameters from earlier studies of CO hydrogenation on unpromoted Fe and Fe/K catalysts and the conditions for which they were determined. Note that data for Fe/Cu/K catalysts are not included, since they are chemically more complex and catalytically more active systems — and hence their kinetics are not comparable.

Apparent activation energies for CO hydrogenation on unpromoted Fe catalysts range from 101 to 108 kJ/mol while those for Fe/K catalysts range from 92 to 101 kJ/mol. Reaction orders for H₂ and CO on unpromoted Fe catalysts lie in ranges of 0.8 to 1.4 and –0.4 to 0.5, respectively. For Fe/K catalysts H₂ and CO orders span ranges of 0.6 to 0.9 and –0.2 to 0.2, respectively. Thus, activation energies for both catalysts span a relatively narrow range of values, while reaction orders for Fe span a wide range and for Fe/K a relatively narrow range.

It is apparent from these data (Table 8) that effects of H₂ and CO partial pressures on the reaction rate may be represented adequately by an SPLE over a narrow range of temperature and concentration conditions, but predictions of rate cannot be extrapolated outside this range with any confidence. In addition, this simplistic rate function does not account for changes in concentration dependence with temperature or effects of product inhibition (e.g., by water) or of promoters on rate. Striking differences in CO reaction orders are particularly evident between our previous

Table 8

Parameters for rate expressions of the form $-r_{\text{CO}} = k (P_{\text{H}_2})^X (P_{\text{CO}})^Y$ for the Fischer–Tropsch reaction on iron catalysts

Type of catalyst (reactor)	K/Fe (atom)	Temperature (°C)	Partial pressures ^a		X (±)	Y (±)	E _{act} ^b	Ref.
			H ₂ (atm)	CO (atm)				
Fe powder (differential)	0	200–230	0.25–0.75	0.25–0.50	1.2	0.54	101	[15]
Fe powder on monolith, (recycle)	0	200–240	0.9–1.4	0.4–0.9	1.4	–0.42	101	This work
Fe/K powder on monolith, (recycle)	0.013	200–240	0.9–1.4	0.4–0.9	0.60	–0.05	92	This work
Fused Fe (differential)	^d	240	7–12	1–8				[16]
15% Fe/Al ₂ O ₃ (micro) ^c	0	275	0.6–0.9	0.1–0.4	1.14 (0.1)	–0.05(0.07)	108	[17]
Fe ₂ O ₃ (slurry)	0	207	1–8	1–8	1.07	–0.51		[20]
K/Fe ₂ O ₃ (slurry)	0.011	240	1–8	1–8	0.91	–0.18		[20]
Precip. Fe (slurry)	0	213–253	1.6–9.0	1.3–7.0	1.0 (0.2)	0.2(0.2)		[21]
Precip. K/Fe (slurry)	0.016	213–253	1.6–9.0	1.3–7.0	0.8 (0.2)			[21]
Precip. K/Fe (slurry)	0.0008	213–253	1.6–9.0	1.3–7.0		0.2(0.2)		[21]
Precip. K/Fe (slurry)	0.0016	213–253	1.6–9.0	1.3–7.0		0.1(0.2)		[21]
Precip. K/Fe (slurry)	0.0049	213–253	1.6–9.0	1.3–7.0		0.1(0.2)		[21]

^a Actual partial pressures in the reactor.^b Apparent activation energy (kJ/mole) based on total CO converted.^c Methane production only.^d Promoted with K₂O, Al₂O₃ and SiO₂.

low-pressure [15] and the subject high-pressure work i.e., CO reaction orders of 0.54 and –0.42 for low and high P_{CO} , respectively. This emphasizes the need for a more general, shifting-order rate expression, e.g., a Langmuir–Hinshelwood expression based on a sequence of elementary steps, provided that the steps and assumptions are chosen carefully.

The previously reported rate expression of Atwood and Bennett for CO hydrogenation on a fused Fe/K catalyst [18], $-r_{\text{CO}} = k P_{\text{H}_2} P_{\text{CO}} / (P_{\text{CO}} + k P_{\text{H}_2} \text{O})$, is an example of a simple Langmuir-type expression which provides only an approximate fit of the data for Fe/K catalysts listed in Table 8, i.e., it predicts simple orders for H₂ and CO of 1.0 and 0 respectively. To our knowledge, no similar shifting-order rate expressions are reported for unpromoted Fe.

4.1.2. Kinetic model for Fischer–Tropsch synthesis on Fe

A robust, mechanistically-based model for FTS on iron, developed by Rankin [24], is considered here. The Fischer–Tropsch reaction is represented by 11 elementary steps shown in Table 9 where (g) and (a) refer to gas-phase and adsorbed species and s represents an empty surface site. CO adsorption and subsequent dissociation (Steps 1 and 2) have been shown to occur rapidly in a near- or quasi-equilibrium on iron. Hydrogen adsorbs dissociatively and likewise rapidly in quasi-equilibrium (Step 3). CH₂ species are formed

Table 9

Elementary steps in the Fischer–Tropsch reaction on iron [24]

1.	CO (g) + s ⇌ CO (a)
2.	CO (a) + s ⇌ C(a) + O (a)
3.	H ₂ (g) + 2s ⇌ 2H (a)
4.	C (a) + H (a) ⇌ CH (a) + s
5.	CH (a) + H (a) ⇌ CH ₂ (a) + s
6.	CH ₂ (a) + H (a) ⇌ CH ₃ (a) + s
7.	O (a) + H ₂ (g) → H ₂ O(a)
8.	O (a) + CO (g) → CO ₂ (a)
9.	C _{n-1} H _{2n-1} (a) + CH ₂ (a) $\xrightarrow{\text{rds}}$ C _n H _{2n+1} (a) + s
10.	C _n H _{2n+1} (a) + H (a) $\xrightarrow{\text{rds}}$ C _n H _{2n+2} (g) + 2s
11.	C _n H _{2n+1} (a) + S $\xrightarrow{\text{rds}}$ C _n H _{2n} (g) + H (a) + s

through the stepwise addition of adsorbed H to adsorbed carbon (Steps 4–6). Oxygen is eliminated as either H₂O or CO₂ by reaction of an adsorbed oxygen with H₂ (Step 7) or CO (Step 8). Since the later step is generally preferred on Fe, it is assumed that no oxygenated hydrocarbons are produced. Propagation of hydrocarbons (Step 9) is assumed to occur by the addition of a CH₂ species to a growing or propagating chain. Termination of chain growth is represented by either addition of a hydrogen to form an alkane (Step 10) or by elimination of the β-hydrogen to form an α-olefin (Step 11).

The following assumptions are implicit in the model: (1) adsorbed CO is the most abundant surface intermediate and (2) the propagation and termination

steps (Steps 9–11) are slow relative to the other steps. In this respect, they are co-rate determining steps.

The model is developed by applying a steady-state approximation for an adsorbed hydrocarbon of carbon number n such that the rate of production equals the rate of depletion. The resulting expression, in terms of methane turnover number (N_1), is

$$N_1 = \frac{k' (k_7 P_{H_2} + k_8 P_{CO})^{1/3} (P_{H_2})^{7/6} (P_{CO})^{1/3}}{(1 + K_1 P_{CO})^{5/3}} \quad (5)$$

where

$$k' = \left(\frac{k_{10}^6 K_6^4 K_5^2 K_4^2 K_3^7 K_2^2 K_1^2}{k_9^2} \right)^{1/6} \quad (6)$$

Turnover frequencies for hydrocarbons with carbon numbers > 1 are determined by multiplying the turnover frequency at the next lower carbon number by the chain propagation probability, α :

$$N_i = \alpha N_{i-1} \left(\frac{k_{10, i}}{k_{10, i-1}} \right) \quad (7)$$

where $i = n$ is the index for the hydrocarbon of n carbon atoms and the term in parentheses is equal to unity if propagation is independent of chain length. In terms of the elementary steps in Table 9, α can be expressed as the rate of propagation to form a higher hydrocarbon over the sum of the propagation and termination to paraffin and α -olefin rates, represented symbolically by

$$\alpha = \left(\frac{k_9 \theta_{CH_2}}{k_9 \theta_{CH_2} + k_{10} \theta_H + k_{11} \theta_s} \right) \quad (8)$$

Where θ_i = the surface coverage of the species 'i'.

Values for K_1 and k' calculated from the data obtained in this study are 0.26 and 0.027 for unpromoted catalysts at 220°C; the corresponding values for the K-promoted catalysts are 1.48 and 0.026. Using these values in Eq. (5) to calculate methane turnover frequencies gives methane formation rates an order of magnitude higher than the measured rates. From Eq. (5), apparent orders of reaction calculated for the power-law expression can vary from 1.1 to 1.5 for hydrogen and from -1.3 to $+0.67$ for carbon monoxide. Thus, nearly all the data from this and previous studies (Table 5) for unpromoted iron fall within these ranges, including the low pressure data from our previous study showing a CO order of 0.54. Nevertheless,

the hydrogen orders of $+0.60$ to 0.9 for Fe/K catalysts (Table 5) are significantly lower than the values predicted by Eq. (5), indicating that the proposed sequence of elementary steps may not be appropriate for the FT reaction on K-promoted iron. Since potassium is known to decrease the hydrogenation capacity of iron catalysts, the hydrogenation to form an alkane termination step (Eq. (10), Table 6) would play a lesser role in the production of total hydrocarbons on Fe/K relative to Fe.

4.2. Deactivation kinetics

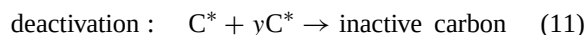
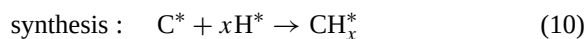
Other than our previous study at 1 atm, there are no previously reported quantitative studies of deactivation kinetics for CO hydrogenation on Fe or Fe/K catalysts, although Pijolat et al. [10] and Jung and Thomson [13] did report a few kinetic data showing changes in activity during CO hydrogenation on Fe/Al₂O₃ as a function of temperature and time. The results of this work shows that rates of deactivation of Fe and Fe/K catalysts are very significant at reaction temperatures exceeding 250°C, reducing normalized activity to less than 50% of the initial activity within 6–80 h. The dependence of deactivation rate on temperature is large for both Fe and Fe/K catalysts; indeed activation energies are on the order 60–100 kJ/mol. Qualitatively, these results are in good agreement with those observed by Pijolat et al. [10] showing 40–85% loss of CO hydrogenation activity for Fe/Al₂O₃ over 20 h at 225–275°C. Initial reaction rates for both unpromoted (Fig. 6) and K-promoted catalysts generally increase as reaction temperature increases. However, measurements of the activity after 24 h on stream shows that for both catalysts, the highest rates are at the lowest temperature of 250°C, with rates decreasing as temperature increases.

The results of this study indicate that, while initial activities depend strongly on CO and H₂ concentrations, normalized activities are relatively independent of reactant concentration. Rather, deactivation rate depends principally on the concentration of remaining active sites and thus is first order in normalized activity.

Since in this study, precautions were taken to eliminate sulfur impurities in the feed, the observed deactivation at higher reaction temperatures was

clearly not a result of sulfur poisoning. Surface area measurements before and after reaction [15] indicate that sintering was not responsible for lost of activity. Moreover, our companion studies involving temperature-programmed hydrogenation of carbonaceous species in the used catalysts [14] and Mössbauer spectra indicate that during high temperature reaction, relatively reactive carbons and carbides are transformed to less reactive forms. Accordingly, the observed losses of activity in this work can be attributed to transformation of active carbons to inactive carbons and of active carbides to less active carbides.

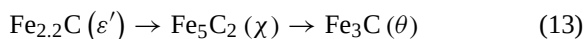
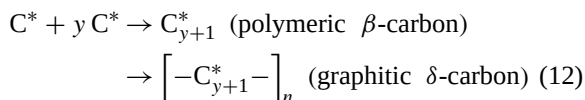
Niemantsverdriet et al. [25] have provided support for a competition model in which several reaction pathways (Eqs. (9)–(11)) compete for the carbon deposited on the surface by the dissociation of CO.



The activity of supported iron catalysts during the first few minutes and hours of reaction has been shown to track the formation of carbides [13,26]. Raupp and Delgass [26] concluded that the bulk iron acts as a getter for carbon and as such, controls the surface composition. For the low-pressure reaction experiments reported earlier [15], a period of 18 hours was required for the catalyst to reach its maximum steady-state activity. Carburization of this low surface area ($10 \text{ m}^2/\text{g}$), unsupported catalyst (dispersion less than 0.5%) may not be complete before the 18 h. At ten atmospheres, the time to reach steady-state is reduced to six hours or less. This is probably due, at least in part, to an increased CO partial pressure, leading to higher rates of CO adsorption and dissociation on the surface and thus provide a larger pool of carbon available to form bulk carbides. In addition, surface areas of the unpromoted catalysts measured by hydrogen chemisorption were higher in this study (70 relative to $30 \mu\text{moles/g}$ of catalyst) due to the alumina added in this work. This higher surface area indicates smaller iron crystallites, which translate to an effective decrease in the distance carbon has to diffuse to reach the crystallite interior during carbide formation. Both effects (CO partial pressure and crystallite size) probably contributed to the observed decrease in time to reach steady-state.

Under mild steady-state reaction conditions, Eq. (11) is not a serious competitor for carbon atoms. As reaction temperatures increase, however, it starts to play a significant role. While formation of hydrocarbon products is increased initially over lower temperatures, the effect of significantly less active carbon species becomes evident immediately. Activity decreases very rapidly, especially at the higher temperatures. For example, a 40% drop in activity is observed within 2 h at 280°C on the unpromoted catalyst.

We propose the following two-step, parallel model for loss of activity during CO hydrogenation on Fe at high temperature:



Formation of a polymeric carbon (Eq. (12)) has been reported during CO hydrogenation on nickel and ruthenium catalysts [11,27–30]. Our previous TPD study [14] provides evidence for transformation of polymeric β -carbon to graphitic δ -carbon during deactivation of the same catalysts. Recent XRD and TEM evidence obtained in the characterization of unsupported Fe [13] and slurry-phase iron catalysts [31] suggests that the active catalytic phase in FTS may be ϵ' -carbide ($\text{Fe}_{2.2}\text{C}$) which over long periods (hours to days) of reaction converts to the χ -carbide (Fe_5C_2) of lower carbon content. Formation of θ -carbide (Fe_3C) observed in a used Fe/ZSM-5 FT catalyst has been linked to loss of catalytic activity [32]. Eq. (13) is a reasonable scenario, considering that the order of increasing thermal stability for carbides is $\epsilon' < \chi < \theta$ [33]; presumably Fe–C bond strength increases in the same order. Thus, we predict (for the same order of carbides) decreasing activity for hydrogenation of the surface atomic carbon, i.e. Fe–C, to a CH_a intermediate (Step 4 in Table 6). Accordingly we propose parallel paths for deactivation of FT catalysts involving simultaneous conversion of atomic to polymeric to graphitic carbon and of active carbon-rich carbides to inactive carbon-poor carbides.

However, these two deactivation paths, while occurring in parallel, may be coupled, since carbon atoms would be precipitated in the transformation of carbon-rich ϵ' to progressively more iron-rich χ and θ

phases. Jung and Thomson [13] have speculated that carbon atoms precipitated by the ϵ' to χ transformation may serve as nucleation sites for the Boudouard reaction. We hypothesize that these carbon atoms formed in carbide transformations (Eq. (13)) may serve as nucleation sites for polymerization to β -carbon and/or crystallization to graphitic δ -carbon (Eq. (12)).

In this study, activation energies for deactivation of 108 and 58 kJ/mol were observed for Fe and Fe/K catalysts, respectively. In previous studies of nickel deactivation, activation energies of 32 kJ/mole [27] and 30 kJ/mole [28] were reported for the transformation of surface α -carbon to a polymeric β -carbon. In studies of deactivation of ruthenium catalysts during CO hydrogenation, Bowman and Bartholomew [29] found an activation energy of 40 kJ/mol, while Mukkavilli et al. [30] reported a value of 69 kJ/mole for the α to β transition. Although this latter value (69) is close to the value of 58 found in our study for CO hydrogenation on Fe/K, the much larger value of 108 kJ/mol observed for CO hydrogenation on Fe suggests that a different mechanism may be responsible for deactivation of iron catalysts, e.g., transformation of polymeric to graphitic carbon and/or of active carbides to inactive carbides. To our knowledge, no activation energy data are available for transformation of polymeric β -carbon to graphitic δ -carbon or for carbide transformations.

In concluding the discussion of the kinetics of Fe catalyst deactivation in FT synthesis it would be appropriate to discuss how the present work, which focuses on deactivation by carbon, fits in the broader context of deactivation problems encountered in FT synthesis processes. These problems may include catalyst poisoning, deactivation by carbon, transformations of active carbides to inactive forms, oxidation by water, and mechanical failure of catalyst particles. The present work has application quantitatively to processes in which poisons have been removed and to those regions in the reactor in which partial pressures of CO and H_2 are moderately high, while partial pressures of CO_2 and H_2O are low. These regions are more likely to occur when CO conversion is low, carburizing conditions are favored, and/or the catalyst has low synthesis activity. In reaction regimes involving high P_{CO_2} and P_{H_2O} (which are more likely to occur at high conversions either in a slurry reactor or near the exit of a fixed bed reactor), iron catalysts are more likely to deactivate due to oxidation of iron metal/iron carbides to

iron oxides. Nevertheless, the present work has qualitative application to processes involving high conversions and high partial pressures of H_2O and H_2 , since carbon formation and transformation of carbides are also observed commercially under these conditions, albeit they occur much more slowly than observed in this study. While it would be desirable to investigate the kinetics of deactivation by carbon deposition and carbide transformations at high conversions and high P_{H_2O} , it is not practical in view of the substantially lower rates as a result of high rates of carbon atom gasification relative to deposition. While it would be highly desirable to determine quantitatively the effects of P_{H_2O} , P_{CO} , and P_{H_2} on the kinetics of deactivation by carbon, this was not within the scope of the present study. Moreover, this study did not address qualitative effects of H_2O . Effects of P_{CO} , and P_{H_2} were addressed qualitatively (as discussed in Section 4.3).

The catalysts used in this study contained mostly 95+ % Fe and were relatively inactive for the water–gas-shift (WGS) reaction over the range of conditions studied; accordingly, the partial pressure of H_2 was not substantially increased by the WGS reaction and the carbon-forming potential was relatively high. By comparison, commercial precipitated Fe/Cu/silica catalysts contain large fractions of iron oxides and are relatively more active for the WGS, they generate substantial H_2 via this reaction, and their carbon-forming potential is comparatively lower.

4.3. Effects of changes in reactant partial pressures and of potassium on deactivation rate

Effects of carbon monoxide concentration on initial activities and deactivation rates of Fe and Fe/K, while secondary to the effects of site concentration and temperature, are nevertheless significant. CO and H_2 appear to have opposite effects on the activity of the unpromoted catalyst (Fig. 4). In this study we observed that maximum reaction rates (both initial and steady-state) on Fe are obtained with a higher hydrogen partial pressure (1.5 atm) and a lower CO partial pressure (0.5 atm). Increasing the CO pressure to 1.0 atm significantly decreases the rates of CO and H_2 consumption. Similarly deactivation rates increase with increasing P_{CO} and decreasing P_{H_2} .

This is in contrast to the effect of CO on the potassium-promoted catalyst (Fig. 5) for which increasing CO partial pressure leads to increased rates, the maximum rates being obtained at the higher H₂ pressure (1.5 atm) and the higher CO pressure (1.0 atm). This increase in both CO and H₂ consumption rates with increased CO partial pressure is consistent with kinetic data presented above for FT synthesis on the K-promoted catalyst over the temperature range 200–240°C.

It has been shown that synthesis rates over iron catalysts are enhanced in the presence of potassium [34]. Indeed, comparison of reaction rate versus time data obtained at hydrogen and CO partial pressures of 1.5 and 1.0 atm, respectively, and at a temperature of 260°C (Figs. 4 and 5) verifies this point. On the unpromoted catalyst, the hydrogen consumption rate at time zero is about 750 $\mu\text{mole/g-min}$, decreasing to 250 $\mu\text{mole/g-min}$ after 24 h. Corresponding rates on the potassium-promoted catalyst are 1400 and 1000 $\mu\text{mole/g-min}$, respectively. Potassium is thought to increase the strength of CO adsorption on the catalyst surface and to weaken the H₂ adsorption strength [19,35], leading to an increased ratio of CO to hydrogen on the surface [36]. Reaction rates will be enhanced if the effect of the weakened C–O bond is not outweighed by the effect of the diminished level of adsorbed hydrogen.

5. Conclusions

1. Rate data for Fischer–Tropsch synthesis on both Fe and Fe/K catalysts in unsupported form were obtained at 10 atm total pressure, 200–240°C, and partial pressures of H₂ and CO of 0.9–1.4 and 0.4–0.9 atm, respectively. Reaction orders based on a simple power-law expression (SPLE) with respect to H₂ and CO are 1.4 and –0.42 for the unpromoted Fe catalyst and 0.60 and –0.05 for the K-promoted Fe catalyst. Apparent activation energies for reaction are 101 kJ/mole (Fe) and 92 kJ/mole (Fe/K). The SPLE provides a good fit of the kinetic data over a narrow range of reaction conditions.
2. A more general, mechanistically-based rate expression was derived from a sequence of elementary steps, including dissociative adsorp-

tion of H₂, molecular adsorption of CO followed by dissociation to C and O atoms, hydrogenation of atomic carbon to CH_x species, polymerization of CH_x species to hydrocarbons, and formation of CO₂ from CO and O. Rate data for Fe are fitted well by this expression, while the data for Fe/K are not.

3. Deactivation rate data were obtained during Fischer–Tropsch synthesis on the same Fe and Fe/K catalysts at 10 atm total pressure, 250–280°C, and partial pressures of H₂ and CO of 0.6–1.4 and 0.3–0.9 atm, respectively, for periods of 24–65 h; from a fit of these data to a generalized power-law expression (GPPE) of the form $r_d = k_d'(a - a_\infty)^m$, values of the deactivation order m are found to be unity for both catalysts. Activation energies for deactivation are 108 and 58 kJ/mole for Fe and Fe/K catalysts, respectively.
4. A deactivation model involving parallel paths is proposed, i.e. simultaneous conversion of (a) atomic to polymeric to graphitic carbon and (b) active carbon-rich carbides to inactive carbon-poor carbides. The activation energy for deactivation of Fe/K is consistent with conversion of atomic to polymeric carbon being rate determining, while that for Fe may involve conversion of either polymeric to graphitic carbon and/or of active carbides to inactive carbides as the rate-determining step.
5. The Fe and Fe/K catalysts of this study have selectivities for hydrocarbons (i.e., 80–90 at.% C from CO) that are higher and CO₂ selectivities (10–20 at.% C) lower than typical Fe FT catalysts, probably due to their high iron and low oxide contents. CO₂ and H₂O selectivities were also relatively low in this study as a result of operation at low CO conversions. Values of the propagation probability are lower for Fe (i.e., 0.63 to 0.76) and about the same for Fe/K (i.e., 0.72 to 0.89) relative to those typically reported for Fe FT catalysts. Changes in product selectivity with time at higher reaction temperatures are relatively modest (1–20%) over a period of 60 hours, while activity losses during this same period are much larger (i.e., 40–80%); selectivities for CO₂ are constant and values of α decrease only 5–6% during this reaction period.

Acknowledgements

The authors acknowledge financial support of this work by the Department of Energy, Pittsburgh Energy Technology Center and Brigham Young University. The authors are grateful for technical assistance by Michael Bayles, John Eng, Byron Johnson, and Matthew Stoker. The development of the kinetic model by Jeffrey L. Rankin is also gratefully acknowledged.

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